

LILLEHAMMER-SAETHERENGEN, Norway

WMO: 013780

Lat: 61.093N

Lon: 10.477E

Elev: 239

StdP: 98.49

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.0	-16.9	-21.7	0.6	-18.7	-19.5	0.7	-16.6	7.6	1.1	6.2	2.1	1.6	10	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	26.4	16.3	24.2	15.6	22.1	14.8	17.7	23.1	16.8	21.5	15.9	20.4	2.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.6	18.9	15.0	10.9	18.4	14.0	10.3	17.6	50.7	23.1	47.9	21.4	45.3	20.4	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.3	5.3	4.5	-20.6	28.6	3.0	2.0	-22.8	30.1	-24.5	31.2	-26.2	32.3	-28.4	33.8
			-20.9	19.2	3.0	1.2	-23.0	20.1	-24.7	20.8	-26.4	21.4	-28.5	22.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-6.9	-5.4	-0.8	4.6	9.7	14.1	16.5	15.0	10.7	4.2	-1.1	-5.4
	DBStd	9.11	5.44	5.11	4.50	3.21	3.96	3.01	3.01	2.52	3.00	3.66	4.27	5.67
	HDD10.0	2529	523	431	334	165	54	3	0	1	27	183	332	478
	HDD18.3	5018	782	664	592	413	268	133	74	107	228	438	582	736
	CDD10.0	582	0	0	0	2	45	124	203	155	49	3	0	0
	CDD18.3	29	0	0	0	0	1	5	19	3	0	0	0	0
	CDH23.3	261	0	0	0	0	21	61	155	24	0	0	0	0
	CDH26.7	33	0	0	0	0	1	6	26	1	0	0	0	0
(14) Wind	WSAvg	1.9	1.7	1.7	1.9	2.0	2.2	2.2	1.9	1.8	1.8	1.8	1.8	1.9
(15) Precipitation	PrecAvg	826	61	42	43	46	75	78	101	104	69	82	88	59
	PrecMax	979	126	106	97	79	150	158	215	160	130	247	313	97
	PrecMin	585	18	13	1	14	4	7	16	26	31	16	21	23
	PrecStd	105	29	23	26	20	38	41	42	41	28	53	62	22
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.5	6.8	11.8	18.1	26.1	27.6	28.9	25.6	20.6	15.7	10.8	7.2
		MCWB	3.5	3.4	6.1	9.7	14.5	16.0	17.7	16.9	14.0	11.7	8.9	4.7
	2%	DB	3.4	4.3	9.1	15.3	22.3	25.3	27.2	23.7	18.7	12.7	8.1	5.5
		MCWB	1.6	2.2	4.2	8.3	13.1	15.1	17.3	16.3	13.2	9.8	6.3	3.6
	5%	DB	2.0	2.5	6.9	12.7	19.7	22.4	25.1	21.7	17.1	11.1	6.4	4.0
		MCWB	0.7	1.0	2.8	6.8	11.9	14.2	16.5	15.3	12.5	8.6	5.1	2.1
	10%	DB	0.6	1.2	5.0	10.5	16.8	20.1	22.7	19.9	15.6	9.6	4.5	2.3
		MCWB	-0.3	-0.1	1.9	5.5	10.5	12.8	15.6	14.6	11.9	7.6	3.3	1.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	4.1	6.4	10.7	15.7	17.2	19.5	18.3	15.6	12.4	9.1	5.2
		MCDB	5.2	6.1	11.0	16.3	22.4	22.0	26.0	22.9	18.4	14.6	11.0	6.7
	2%	WB	2.0	2.3	4.6	8.8	14.2	16.1	18.3	17.2	14.4	10.4	6.8	3.7
		MCDB	3.1	3.8	8.2	13.8	20.5	22.4	24.3	21.3	17.2	12.2	7.8	5.2
	5%	WB	0.9	1.2	3.3	7.5	12.6	15.1	17.3	16.3	13.4	9.0	5.1	2.2
		MCDB	1.8	2.5	6.2	12.0	17.9	20.9	22.8	20.0	16.0	10.6	6.2	3.6
	10%	WB	-0.3	0.1	2.2	6.1	11.2	14.0	16.4	15.4	12.4	7.9	3.4	1.0
		MCDB	0.6	0.9	4.7	9.8	15.7	19.3	21.3	18.7	14.8	9.4	4.4	2.1
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.5	7.8	8.8	9.3	9.6	9.1	8.6	8.0	5.6	4.3	4.7
		MCDBR	5.9	7.1	10.0	12.9	13.9	13.4	12.5	11.4	10.1	7.6	6.7	7.0
		MCWBR	5.0	5.3	6.2	7.2	6.7	6.4	5.5	5.6	5.8	5.3	5.4	5.9
	5% WB	MCDBR	5.8	6.4	9.4	11.9	11.5	11.4	10.6	9.2	8.6	6.3	6.1	6.4
		MCWBR	5.1	4.8	6.0	6.9	5.9	6.0	5.0	5.0	5.4	5.0	5.3	5.5
(40) Clear-Sky Solar Irradiance	taub		0.225	0.272	0.304	0.344	0.350	0.344	0.361	0.353	0.319	0.303	0.242	0.191
	taud		2.143	2.254	2.252	2.274	2.416	2.472	2.437	2.454	2.532	2.484	2.226	2.026
	Ebn at Noon		530	720	815	847	873	885	857	835	808	691	502	392
	Edh at Noon		36	64	90	108	102	98	99	90	70	52	33	24
(44) All-Sky Solar Radiation	RadAvg		0.28	0.94	2.48	4.09	4.75	5.30	4.80	3.73	2.43	1.08	0.35	0.16
	RadStd		0.05	0.18	0.23	0.43	0.47	0.48	0.61	0.33	0.29	0.19	0.09	0.01

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page